

Arduino-Based MOSFET TERMINAL IDENTIFICATION SYSTEM

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Metal oxide semiconductor field effect transistor (MOSFET) is a semiconductor device widely used for switching applications and amplifying electronic signals. While the facility to determine the type and pin configuration of bipolar junction transistors (BJTs) are available in digital multimeters (DMMs) used in the laboratory, most DMMs do not provide such features for MOSFETs.

This Arduino-based project allows you to identify the type of MOSFET (p-channel or n-channel) and determine the pins that correspond to drain, source and gate terminals of

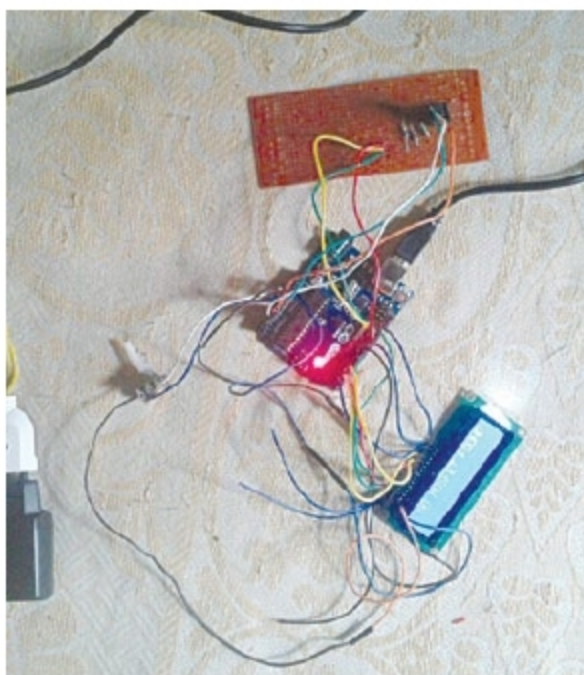


Fig. 1: Authors' prototype

MOSFET. The authors' prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the MOSFET identification system is shown in Fig.

2. The circuit is built around Arduino Uno board, 16x2 LCD (LCD1), three resistors (R1 through R3) and a few other components. MOSFET is a four-terminal device having source (S), gate (G), drain (D) and body (B) terminals.

In most cases, body of the MOSFET is shorted with the source terminal. This makes it operate as a three-terminal device. For n-channel MOSFET (nMOS), D and S are n-type, while terminal B is p-type semiconductor material.

On the other hand, for p-channel MOSFET (pMOS), D and S are p-type, while terminal B is n-type semiconductor material.

As observed from Fig. 2, digital pins 11, 12 and 13 of Arduino UNO are used to apply input to an unknown device (test MOSFET). Whereas, analogue pins A0, A1 and A2

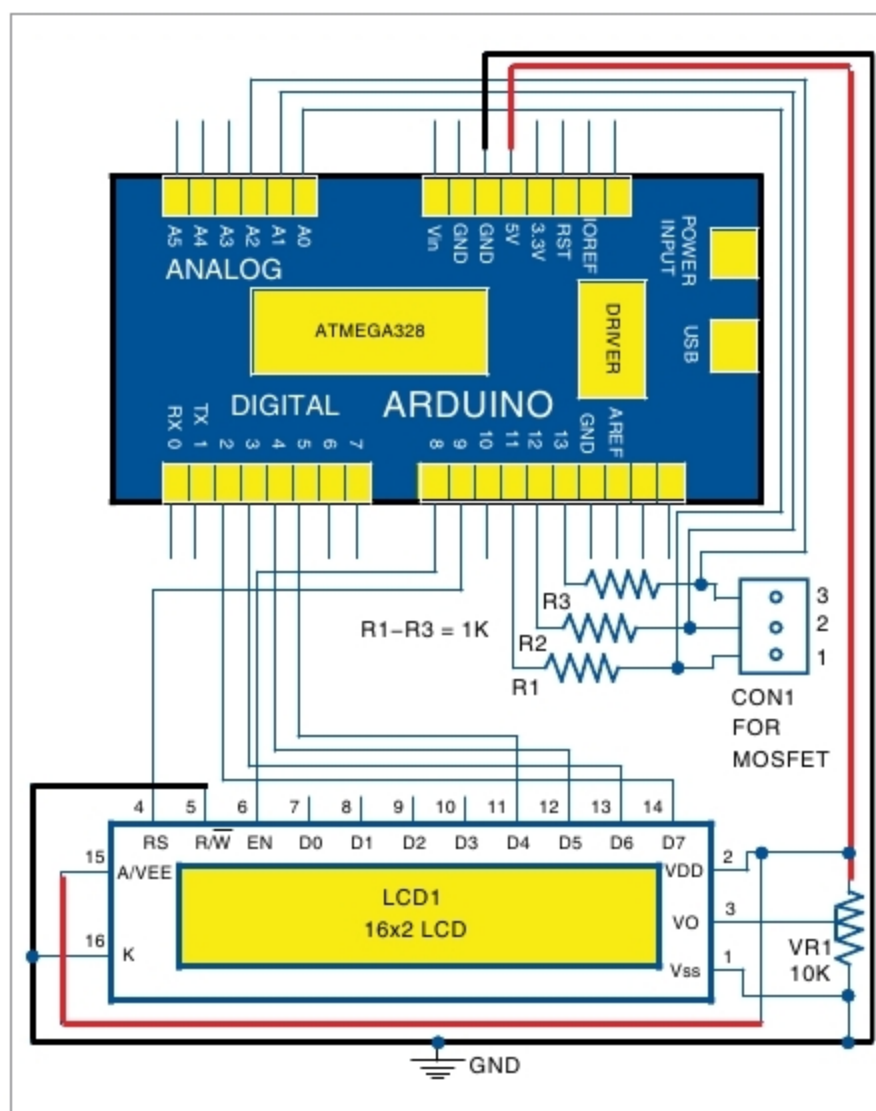


Fig. 2: Circuit diagram of MOSFET identification system